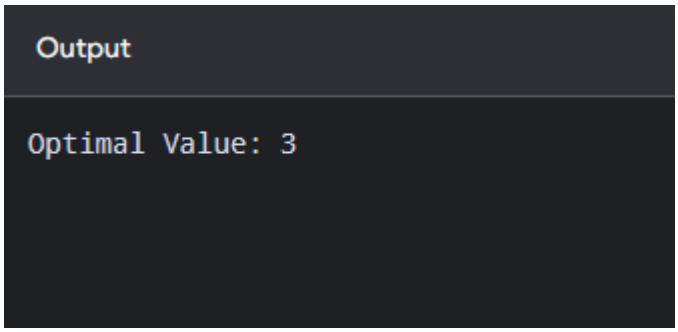
Lab experiment

Minimax Algorithm for Gaming

Coding:

```
# Minimax Algorithm for Gaming

# Game tree
tree = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F', 'G']
}

# Utility values of leaf nodes
values = {
    'D': 3,
    'E': 5,
    'F': 2,
    'G': 9
}

# Minimax function
def minimax(node, is_max):
    if node not in tree:
        return values[node]

    if is_max:
        best = float('-inf')

        for child in tree[node]:
            score = minimax(child, False)
            best = max(best, score)

        return best

    else:
        best = float('inf')

        for child in tree[node]:
            score = minimax(child, True)
            best = min(best, score)
```

```
        return best

# Main program
best_value = minimax('A', True)

print("Optimal Value:", best_value)
```

Output: